Introduction to SQL

SQL:

SQL, or Structured Query Language, is the standard language used to interact with and manage data within a Relational Database Management System (RDBMS). It serves as the primary means of communication between users or applications and the relational database.

SQL is widely supported across various database systems like MySQL, Oracle, PostgreSQL, SQL Server and many others.

SQL is divided into several sublanguages:

- **DDL (Data Definition Language)**
- **DML (Data Manipulation Language)**
- **DCL (Data Control Language)**
- **TCL (Transaction Control Language)**

Role of SQL in DBMS:

- **Data Definition Language (DDL):** SQL provides commands to define and manage the structure of the database. This includes creating, altering, and dropping database objects like tables, views, indexes, and schemas.

Code:

```
CREATE TABLE Employees (
    EmployeeID INT PRIMARY KEY,
    FirstName VARCHAR(50),
    LastName VARCHAR(50)
);
```

- **Data Manipulation Language (DML):** SQL enables users to manipulate the data stored within the database. This involves inserting new data, updating existing data, and deleting data.

Code:

```
    INSERT INTO Employees (EmployeeID, FirstName, LastName) VALUES (1,
'John', 'Doe');
    UPDATE Employees SET FirstName = 'Jane' WHERE EmployeeID = 1;
    DELETE FROM Employees WHERE EmployeeID = 1;
```

- **Data Query Language (DQL):** SQL is most famously used for querying and retrieving data from the database based on specified criteria.

Code:

```
SELECT FirstName, LastName FROM Employees WHERE LastName = 'Doe';
```

- **Data Control Language (DCL):** SQL allows for managing access permissions and security within the database, granting or revoking privileges to users or roles.

Code:

```
GRANT SELECT ON Employees TO user1;
REVOKE DELETE ON Employees FROM user1;
```

- **Transaction Control Language (TCL):** SQL provides commands to manage transactions, ensuring data integrity and consistency by committing or rolling back changes as a single unit of work.

Code:

```
COMMIT;
ROLLBACK;
```

In essence, SQL is the language that allows users and applications to interact with, define, manipulate, and control data within a relational database management system.

Summary: Why SQL is Important in DBMS

Role	Purpose
Schema Definition	Defines structure via DDL
Data Handling	Inserts, updates, deletes, and retrieves data (DML)
Access Control	Grants and restricts user permissions (DCL)
Transaction Management	Ensures consistency and rollback capabilities (TCL)
Querying and Reporting	Powerful querying language for data analysis
Performance Enhancement	Through views, indexes, and optimization features

Why SQL?

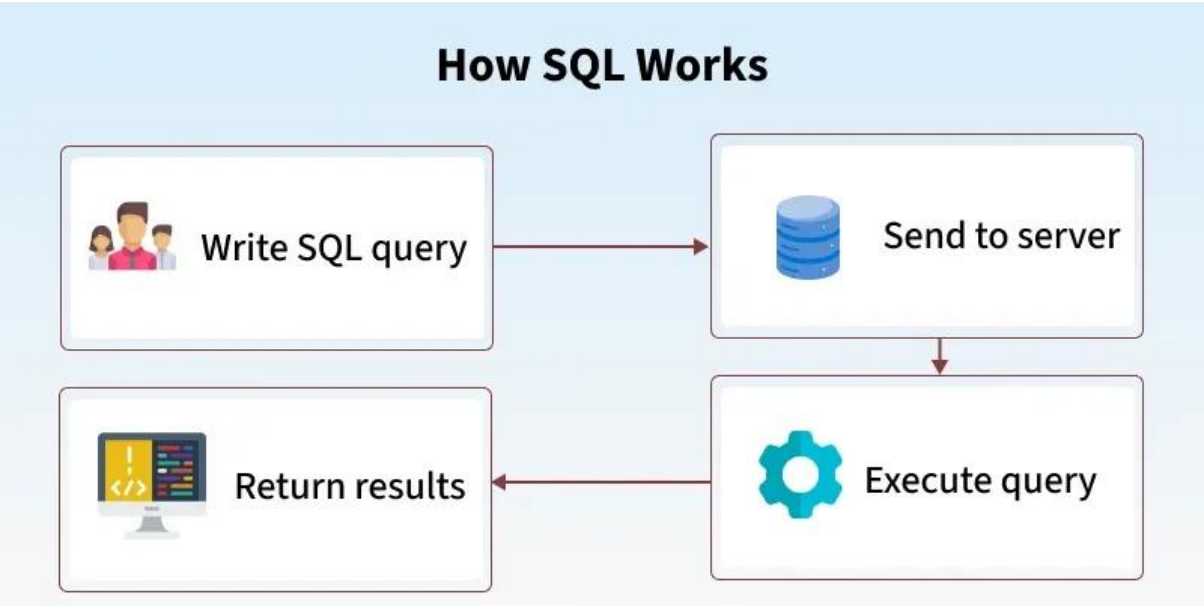
SQL is widely popular because it offers the following advantages −

- Allows users to access data in the relational database management systems.
- Allows users to describe the data.
- Allows users to define the data in a database and manipulate that data.
- Allows to embed within other languages using SQL modules, libraries & pre-compilers.
- Allows users to create and drop databases and tables.
- Allows users to create view, stored procedure, functions in a database.
- Allows users to set permissions on tables, procedures and views.

A Brief History of SQL

- **1970** − Dr. Edgar F. "Ted" Codd of IBM is known as the father of relational databases. He described a relational model for databases.
- **1974** − Structured Query Language (SQL) appeared.
- **1978** − IBM worked to develop Codd's ideas and released a product named System/R.
- **1986** − IBM developed the first prototype of relational database and standardized by ANSI. The first relational database was released by Relational Software which later came to be known as Oracle.
- **1987** − SQL became the part of the International Organization for Standardization (ISO).

How SQL work?



When you interact with a database, you commonly use SQL commands to request operations like retrieving or modifying data. These commands are processed by the SQL Engine, which breaks down, optimizes and executes them efficiently. The SQL Engine ensures the database operates smoothly, handling all data retrieval and updates.

While DBMS tools like MySQL and SQL Server provide interfaces to write SQL queries. These tools provide a user-friendly way to write and execute SQL queries, but internally they rely on their respective SQL Engines to process these commands.

For example, MySQL uses its own SQL Engine to parse, optimize and execute queries, while SQL Server has a different SQL Engine for the same task. These engines ensure that SQL queries are executed in a way that respects the underlying database structure and the specific DBMS optimizations.